

Money and Success in Football

DOES FINANCIAL STRENGTH PREDICT LEAGUE POSITION?

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Does Financial Strength Predict League Position?

Research Question: Does a squad's financial rank within its league systematically determine where it finishes in the final standings?

Why this matters:

- Europe's five major leagues generate over **€18 billion** in annual revenue; transfer spending exceeded **€12 billion** in 2023–24 alone.
- UEFA introduced **Financial Fair Play** precisely to limit financial dominance — implying money *does* buy success.
- If outcomes are financially predetermined, competitive drama is fundamentally reduced for fans of smaller clubs.

This study uses 975 team-seasons from 5 leagues and 10 seasons (2015–16 to 2024–25) to answer with evidence, not opinion.

Two contrasting cases — one financially expected, one statistically miraculous — motivate why the question requires careful empirical investigation.

Two Cases: One Expected, One Miraculous

Leicester City 2015–16

Financial Rank	11th of 20
League	Premier League
Squad value	£161M (avg: £458M)
Model prediction	11th place
Actual finish	1st (Champion)
Residual	– 10.5 positions
Bookmaker odds	5 000-to-1

The largest single-season overperformance in the entire 10-year dataset — a statistical shock that occurs with probability well below 0.1%.

Bayer Leverkusen 2023–24

Financial Rank	2nd (Bundesliga)
Result	Bundesliga champions
Historic feat	First unbeaten Bundesliga season
Context	Ended Bayern's 11-year streak
Residual	–1.8 (within normal range)

Remarkable by football standards, but statistically *expected*: the second-richest squad winning is precisely what the financial hierarchy predicts.

These two cases bracket the full spectrum of outcomes: from the financially-expected champion to the greatest financial upset in modern football.

Data and Scope: 975 Team-Seasons Across Five Leagues

Two data sources:

- **ESPN FC** — final league standings, all five leagues, seasons 2015–16 through 2024–25.
- **Transfermarkt** — squad total market value (€M) at the start of each season.

League	N	Mean MV	$r(\text{FR}, \text{Pos})$
Bundesliga	180	€237M	0.783
La Liga	200	€257M	0.787
Ligue 1	196	€185M	0.791
Premier League	200	€458M	0.801
Serie A	199	€247M	0.889
All	975	€278M	0.813

values: $p < 0.001$.

Key Variable: Financial Rank (FR)

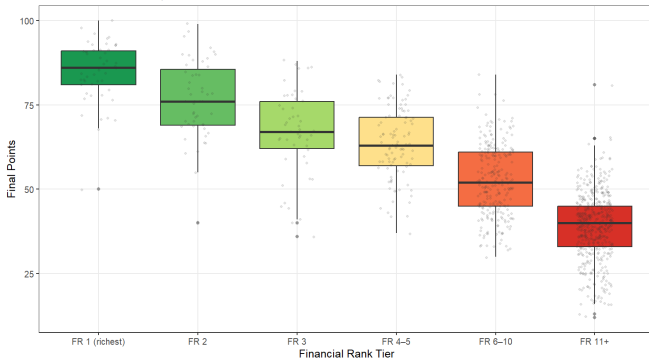
- Squads ranked by market value **within each league-season**: $\text{FR} = 1$ richest, $\text{FR} = n$ least wealthy.
- **Why rank, not raw euros?** A €300M squad is the *richest* in Ligue 1 but only the *fifth-richest* in the Premier League. Ranking within leagues removes this scale problem.
- $\text{FR} = 1$ always means “the wealthiest club in this league this season,” regardless of whether that squad is worth €600M or €150M.

Even before any model: correlations above 0.8 tell us money is a very powerful predictor. Pearson $r = 0.813$, Spearman $\rho = 0.814$; both $p < 0.001$, $N = 975$.

Financial Rank Creates a Balanced, Meaningful Hierarchy

Points Distribution by Financial Rank Tier (2015–2025)

Lower rank number = richer squad • Dots = individual team-seasons



Why Uniformity Matters

- Every league-season contains exactly as many clubs as rank positions, so each FR value appears equally often by construction.
- $FR = 1$ appears exactly 50 times (once per league-season), as do $FR = 2$, $FR = 3$, and so on.
- No structural concentration at any particular rank: the system is **clean and balanced**.

The boxplot confirms a monotonic decline in league points as Financial Rank increases — the

Richer Squads Finish Higher — Across All 975 Team-Seasons

Richer Squads Finish Higher — With Notable Exceptions

$N = 975$ team-seasons • Pearson $r = 0.81$, $p < 0.001$ • Model H quadratic fit • Gold = key story clubs



Financial Rank normalised within each league-season. Axis reversed: top = better.

$$r = 0.813 \quad R^2 = 0.671$$

Pearson correlation
and variance explained

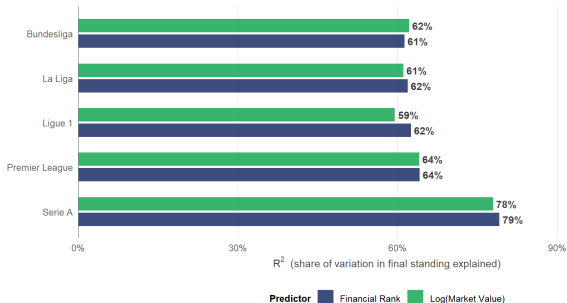
FR vs. Final Position,
 $N = 975$, $p < 0.001$

- FR = 1–2 clubs cluster strongly near the top of their league.
- $FR \geq 15$ clubs cluster near the bottom.
- The fitted curve (Model H) bends at the bottom: diminishing returns.

The Financial Hierarchy Holds in Every League

Money Explains 61–79% of Final League Position

R^2 per league • Both predictors significant ($p < 0.001$) in all leagues



OLS: League_Position ~ predictor, estimated separately per league. Serie A shows the strongest relationship ($R^2 = 0.79$).

League	r	R^2
Bundesliga	0.783	0.613
La Liga	0.787	0.619
Ligue 1	0.791	0.625
Premier League	0.801	0.641
Serie A	0.889	0.791
All (pooled)	0.813	0.671

Financial Rank alone explains 61–79 % of final position *within each individual league*. Serie A shows the strongest relationship ($R^2 = 0.79$); even the weakest (Bundesliga, $R^2 = 0.61$) is very high by 6/14

Model Selection: AIC Identifies Model H as Best Specification

Mdl	Formula	R^2	Adj. R^2	AIC
A	FR only (linear baseline)	0.660	0.660	5099.8
C	Log(Market Value) only	0.533	0.532	5410.6
H	FR + FR² (quadratic)	0.671	0.670	5070.5
H4	FR + FR ² + League + Season	0.671	0.669	5079.3
J	Log(FR)	0.626	0.625	5194.0
E	Champion ~ FR (Logit)	McFadden = 0.548		182.4

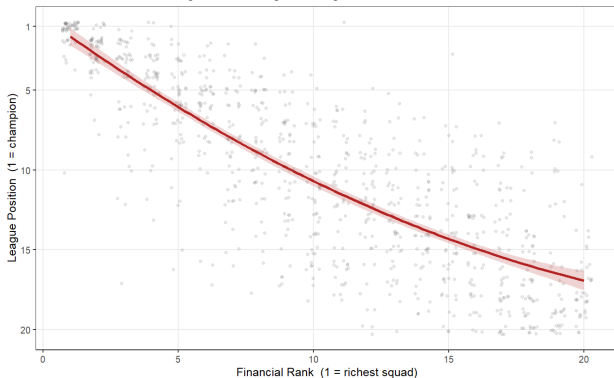
Model H wins on AIC: adding one extra term (FR²) lowers AIC by 29.3 points vs. the linear baseline. Extended Model H4 (league + season controls) scores higher AIC than H — the controls add no explanatory power ($F(5) = 0.247$, $p = 0.941$).

OLS response: League Position (1 = champion), $N = 975$. Lower AIC = better fit per parameter. Bold = preferred specification. Model E uses logistic regression for the binary outcome (champion vs. not).

Model H: Diminishing Returns to Financial Advantage

Quadratic Fit: League Position vs Financial Rank (Model H)

Tests whether the financial advantage shows diminishing returns at high ranks



Model H Equation

$$\widehat{\text{Pos}} = 0.42 + 1.23 \text{FR} - 0.020 \text{FR}^2$$

- $\hat{\beta}_1 = 1.231$ (SE 0.077),
 $p < 0.001$
- $\hat{\beta}_2 = -0.020$ (SE 0.004),
 $p < 0.001$
- $R^2 = 0.671$

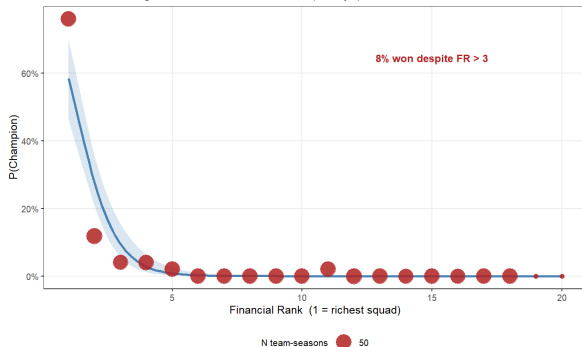
Marginal cost of one rank step

- FR 1 → 2: ≈ 1.19 positions lost
- FR 15 → 16: ≈ 0.63 positions lost

Championship Probability Plummet With Financial Rank

Predicted Probability of Championship vs Financial Rank (Model E)

Blue band = 95% CI of logistic fit • Red dots = observed win rate (sized by N)



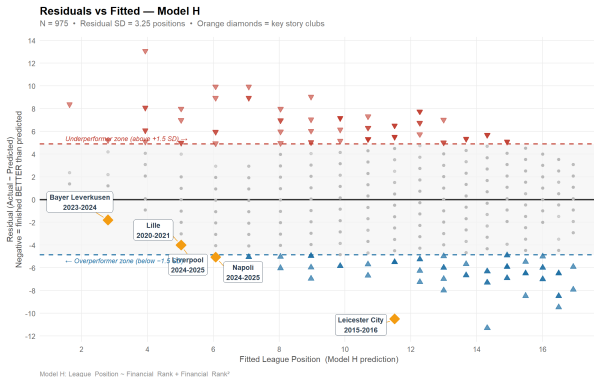
FR	$P(\text{Champion})$
1 (richest)	58.5 %
2	27.8 %
3	9.5 %
5	0.8 %
11	<0.01 %

Model E:

Champion $\sim$ FR (logistic). $\hat{\beta}_{\text{FR}} = -1.298$,
 $p < 0.001$. Each rank step multiplies odds by
0.273 (−72.7 %). McFadden $R^2 = 0.548$.

FR = 1: 58.5 % title probability
— $\approx 3\times$ the baseline for a 20-
team league. This is structural
advantage, not certainty: win-
ning from FR = 11 is virtually
impossible by this model.

Who Beat the Model? Residual Analysis



What Is a Residual?

Residual = Actual - Predicted

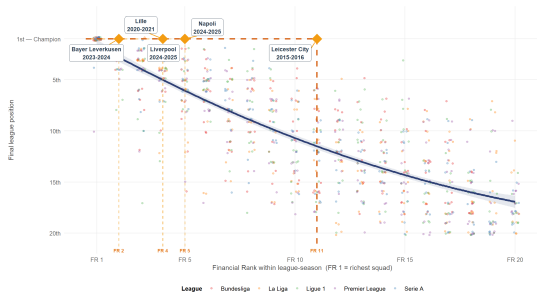
- **Negative:** finished **better** than predicted (**overperformer**)
- **Positive:** finished **worse** than predicted (**underperformer**)
- Near zero: correctly predicted

Dashed lines mark ± 1.5 SD ($\approx \pm 4.9$ positions). Teams outside this band are genuine statistical outliers — performance unexplained by finances alone.

Leicester City 2015–16: The Greatest Statistical Upset

Richer Squads Finish Higher — With Notable Exceptions

N = 975 team-seasons • Pearson $r = 0.81$, $p < 0.001$ • Model fit quadratic fit • Gold = key story clubs



Financial Rank normalised within each league-season. Axis reversed: top = better.

Leicester City 2015–16

Financial Rank	11th of 20 (PL)
Predicted pos.	11.5th
Actual pos.	1st (Champion)
Residual	–10.5 positions
SD units	–3.2 SD
Cook's Distance	Highest in dataset

4 of 50 champions (8%) came from $FR > 3$. Napoli (FR 5), Lille (FR 4), Liverpool (FR 4) were within 1 SD of expectation. Only Leicester (FR = 11) is a genuine statistical outlier.

Robustness Checks: The Core Finding Survives Every Test

Test	Statistic	p -value	Verdict
Quadratic vs. linear (FR^2)	$F = 31.71, 1 \text{ df}$	< 0.001	FR² confirmed
FR $\times$ League interaction	$F = 1.17, 4 \text{ df}$	0.32	Effect uniform
FR $\times$ Season time trend	$F = 0.44, 1 \text{ df}$	0.51	Stable 2015–25
Cubic term (FR^3)	$F = 0.20, 1 \text{ df}$	0.65	Model H sufficient
H4 vs. H (League + Season)	$F(5) = 0.247$	0.941	Controls add nothing
Heteroskedasticity (BP test)	$\chi^2 = 45.5$	< 0.001	Present; HC1 SEs used

Models H2, H3, H4 all score higher AIC than Model H. Neither league fixed effects nor a time trend add explanatory power. Model H ($FR + FR^2$) is the correct and sufficient specification.

Statistical: Under HC1 robust SEs, the FR coefficient remains $p < 0.001$.

Plain English: The finding survives every specification test applied.

Conclusions: What Ten Years of Data Tell Us

1. Money strongly predicts league position.

$r = 0.813$, $R^2 = 0.671$ with FR alone across 975 team-seasons ($p < 0.001$): **67 %** of positional variation explained.

2. The relationship is non-linear (diminishing returns).

FR² term: $F = 31.71$, $p < 0.001$. Marginal penalty: 1.19 positions per rank at the top vs. 0.63 positions at the bottom.

3. The effect is stable across leagues and over time.

FR $\times$ League: $F = 1.17$, $p = 0.32$ (uniform). FR $\times$ Season: $F = 0.44$, $p = 0.51$ (stable 2015–25).

4. Only one true exception in ten years.

4 of 50 champions from FR > 3 . Only Leicester 2015–16 (FR = 11, residual = -10.5 , = -3.2 SD) is a genuine statistical outlier.

Limitations

- Market value as quality proxy (injuries unmodelled)
- Managerial quality not captured
- Panel structure ignored
- Top-five leagues only

Financial dominance is universal, stable, and statistically powerful across all five leagues and all ten seasons. Leicester City 2015–16 is the exception that proves the rule.

Thank You — Questions?

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Does Financial Strength Predict League Position?

$$r = 0.813 \quad R^2 = 0.671 \quad N = 975$$

Five leagues · Ten seasons · One answer: **yes, and very strongly so**

Data: ESPN FC + Transfermarkt

Period: 2015–16 to 2024–25

Models: OLS (A–H, H2–H4), Logit (E, J)

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Key Numbers to Remember

- **76 %** of titles: won by FR = 1
- **92 %** of titles: from financial top 3
- **67 %** of position variance explained by FR alone
- **1** genuine statistical outlier in 10 years: Leicester City 2015–16

Leicester City 2015–16: FR = 11, predicted 11th, won the league. Residual = -10.5 pos. = -3.2 SD. The model said impossible. They proved it — once in a generation.